

ANSWER KEY — Integration Lesson 31 Test: Integration by Parts

II

For the grader · full working shown · give credit for correct method with small arithmetic slips · "+ C" required on indefinite integrals

1. Find $\int x e^{2x} dx$.

Work: parts with $u = x$, $dv = e^{2x} dx \rightarrow du = dx$, $v = e^{2x}/2$. So $\int = x \cdot (e^{2x}/2) - \int e^{2x}/2 dx = (x/2)e^{2x} - e^{2x}/4 + C$. Check by differentiating: $e^{2x}/2 + x e^{2x} - e^{2x}/2 = x e^{2x}$ ✓. **Answer: $(x/2)e^{2x} - (1/4)e^{2x} + C$.**

2. Find $\int 3x \cos x dx$.

Work: pull out the 3, then parts with $u = x$, $dv = \cos x dx \rightarrow du = dx$, $v = \sin x$: $3[x \sin x - \int \sin x dx] = 3[x \sin x + \cos x] + C$. Check: $3 \sin x + 3x \cos x - 3 \sin x = 3x \cos x$ ✓. **Answer: $3x \sin x + 3 \cos x + C$.**

3. u-sub or parts? Then solve: $\int x \sin(x^2) dx$.

Work: u-substitution — the inside function x^2 has fingerprint $2x dx$, and $x dx$ is present. $u = x^2 \rightarrow du = 2x dx \rightarrow \int = (1/2) \int \sin u du = -(1/2) \cos u + C$. Check: derivative of $-(1/2) \cos(x^2)$ is $-(1/2)(-\sin(x^2))(2x) = x \sin(x^2)$ ✓. (Parts would make this worse — always check for the fingerprint first.) **Answer: u-sub; $-(1/2) \cos(x^2) + C$.**

4. Find $\int (x^2 + 1) e^x dx$.

Work: split: $\int x^2 e^x dx + \int e^x dx$. Parts twice on the first: pass 1 ($u = x^2$): $x^2 e^x - 2 \int x e^x dx$; pass 2: $\int x e^x dx = x e^x - e^x$. So $\int x^2 e^x dx = x^2 e^x - 2x e^x + 2e^x$. Add $\int e^x dx = e^x$. Total: $x^2 e^x - 2x e^x + 3e^x + C = e^x(x^2 - 2x + 3) + C$. Check: derivative = $e^x(x^2 - 2x + 3) + e^x(2x - 2) = e^x(x^2 + 1)$ ✓. **Answer: $e^x(x^2 - 2x + 3) + C$.**

5. Find $\int \ln(2x) dx$.

Work: $u = \ln(2x)$, $dv = dx \rightarrow du = (1/(2x)) \cdot 2 dx = (1/x) dx$ (chain rule), $v = x$. So $\int = x \ln(2x) - \int x \cdot (1/x) dx = x \ln(2x) - \int 1 dx = x \ln(2x) - x + C$. Check: $\ln(2x) + x \cdot (1/x) - 1 = \ln(2x)$ ✓. **Answer: $x \ln(2x) - x + C$.**

6. Find $\int x^3 \ln x dx$.

Work: LIATE: Log beats Algebraic, so $u = \ln x$, $dv = x^3 dx \rightarrow du = (1/x) dx$, $v = x^4/4$. $\int = (x^4/4) \ln x - \int (x^4/4)(1/x) dx = (x^4/4) \ln x - \int x^3/4 dx = (x^4/4) \ln x - x^4/16 + C$. Check: $x^3 \ln x + (x^4/4)(1/x) - x^3/4 = x^3 \ln x + x^3/4 - x^3/4 = x^3 \ln x$ ✓. **Answer: $(x^4/4) \ln x - x^4/16 + C$.**

7. Compute $\int_1^{e^2} \ln x \, dx$ exactly.

Work: antiderivative of $\ln x$ is $x \ln x - x$ (parts with $dv = dx$). Evaluate: at $x = e^2$: $e^2 \cdot \ln(e^2) - e^2 = e^2 \cdot 2 - e^2 = e^2$. At $x = 1$: $1 \cdot \ln 1 - 1 = 0 - 1 = -1$. Subtract: $e^2 - (-1) = e^2 + 1 \approx 8.39$. **Answer: $e^2 + 1$.**

8. Compute $\int_0^{\pi/2} x \cos x \, dx$ exactly.

Work: antiderivative (parts, $u = x$): $x \sin x + \cos x$. At $\pi/2$: $(\pi/2) \cdot \sin(\pi/2) + \cos(\pi/2) = \pi/2 \cdot 1 + 0 = \pi/2$. At 0 : $0 \cdot \sin 0 + \cos 0 = 0 + 1 = 1$. Subtract: $\pi/2 - 1 \approx 0.571$. **Answer: $\pi/2 - 1$.**

9. Compute $\int_0^1 x e^x \, dx$ exactly.

Work: antiderivative: $x e^x - e^x$. At 1 : $1 \cdot e - e = 0$. At 0 : $0 - e^0 = -1$. Subtract: $0 - (-1) = 1$. The e 's cancel completely — the area under $y = x e^x$ from 0 to 1 is exactly 1 . **Answer: 1 .**

10. Challenge: $\int x^3 \cos x \, dx$ by the tabular method.

Work: D column (differentiate to 0): $x^3 \rightarrow 3x^2 \rightarrow 6x \rightarrow 6 \rightarrow 0$. I column (antidifferentiate repeatedly): $\cos x \rightarrow \sin x \rightarrow -\cos x \rightarrow -\sin x \rightarrow \cos x$. Diagonal products (each D entry $\times$ the I entry one row lower), signs $+, -, +, -$: $+x^3 \cdot \sin x$, $-3x^2 \cdot (-\cos x) = +3x^2 \cos x$, $+6x \cdot (-\sin x) = -6x \sin x$, $-6 \cdot (\cos x) = -6 \cos x$. Check by differentiating: $(3x^2 \sin x + x^3 \cos x) + (6x \cos x - 3x^2 \sin x) - (6 \sin x + 6x \cos x) + 6 \sin x = x^3 \cos x$ ✓. **Answer: $x^3 \sin x + 3x^2 \cos x - 6x \sin x - 6 \cos x + C$.**